HALS IR Engine Performance Validation

The HALS impulse response engine is compared with ARTA and REW using both a digital driver loopback and an analogue loopback through a MOTU M4 audio interface. These tests assess numerical accuracy, the effects of interface bandwidth limitations, and the effectiveness of interface correction.

Measurements use one 5 sec sweep + a further 5 sec after the sweep at 48 kHz.

Digital loopback without interface correction

Figure 1. Frequency Response

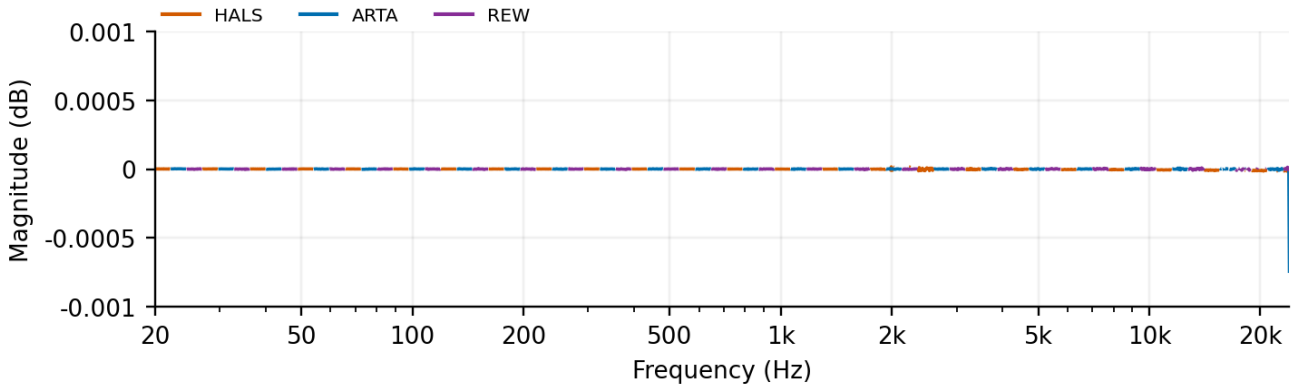


Figure 2. Phase Response

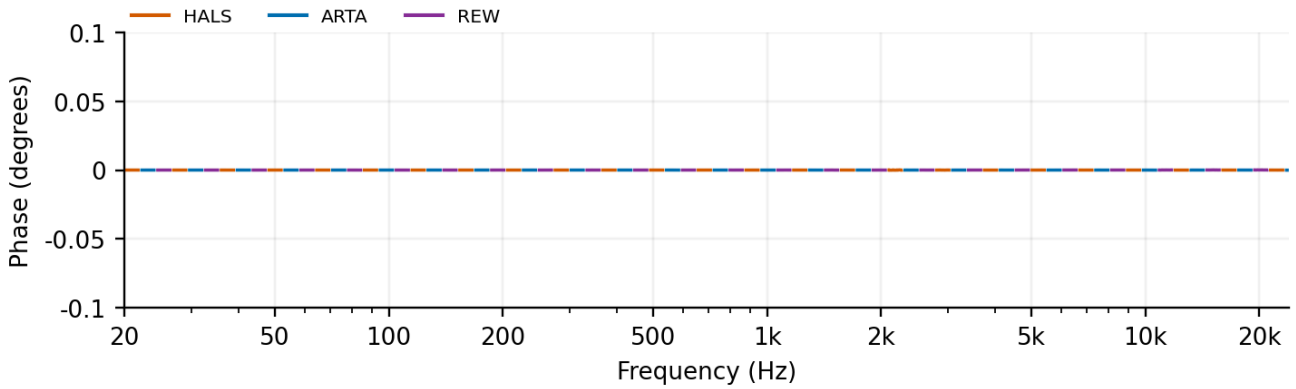


Figure 3. Group Delay

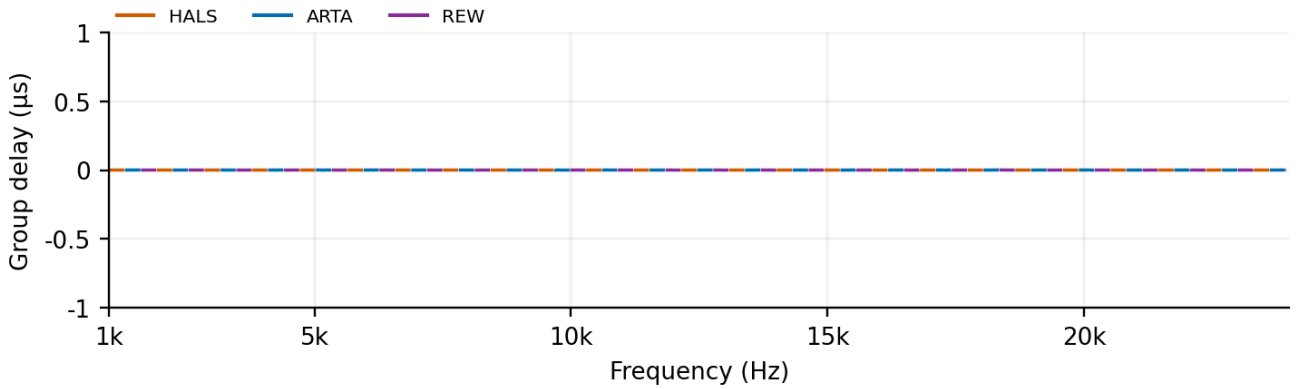
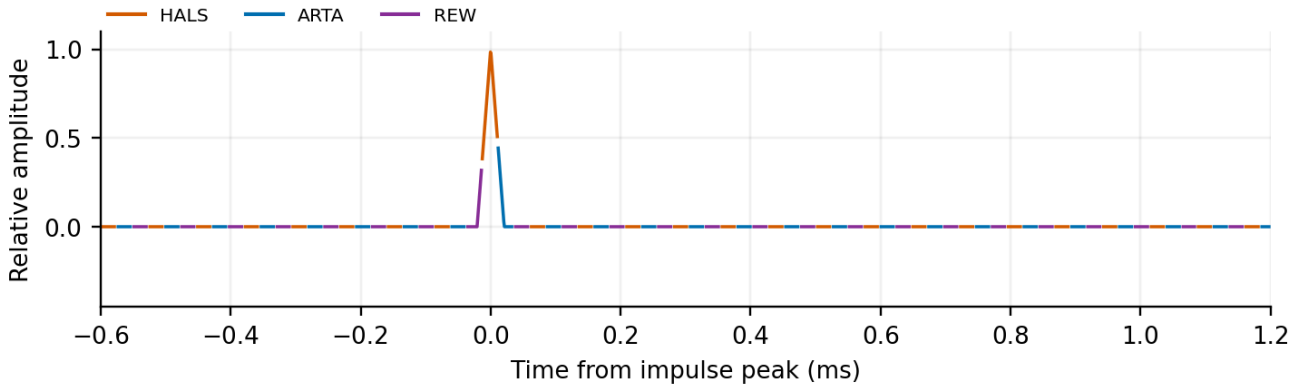


Figure 4. Impulse Response



The frequency response (Figure 1), phase (Figure 2), group delay (Figure 3) and impulse waveform (Figure 4) show the numerical cleanliness of sweep deconvolution without D/A conversion, A/D conversion or an acoustic path. All three engines are effectively transparent. The tiny numerical differences do not represent a meaningful advantage for acoustic measurements.

MOTU M4 hardware loopback without interface correction

The same tests are repeated through the M4's analogue outputs and inputs. Outputs 1/2 are cabled to inputs 1/2, so the recovered response includes the converters, analogue stages and interface filtering.

Figure 5. Frequency Response

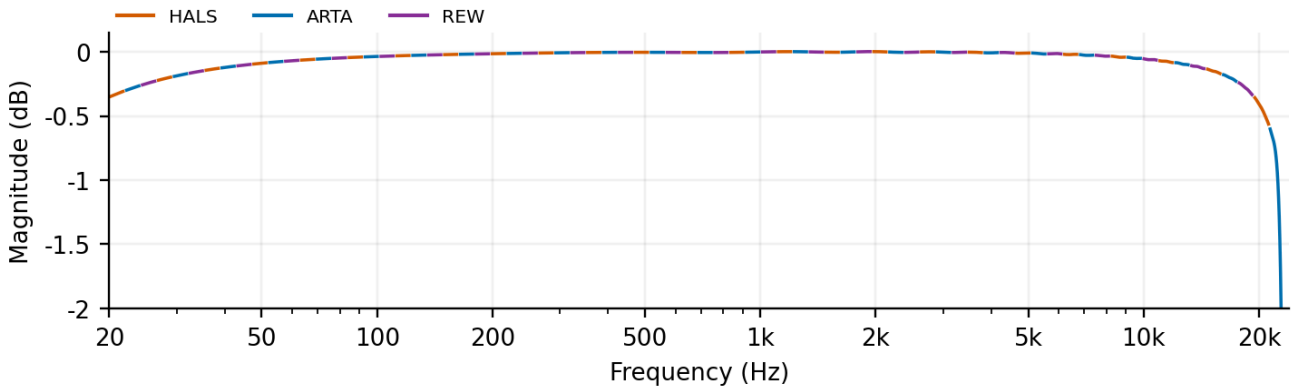


Figure 6. Phase Response

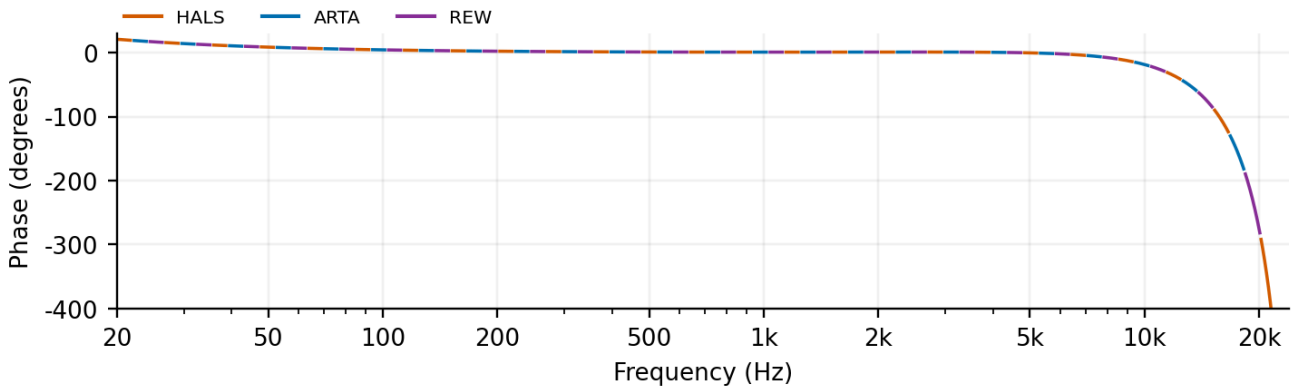


Figure 7. Group Delay

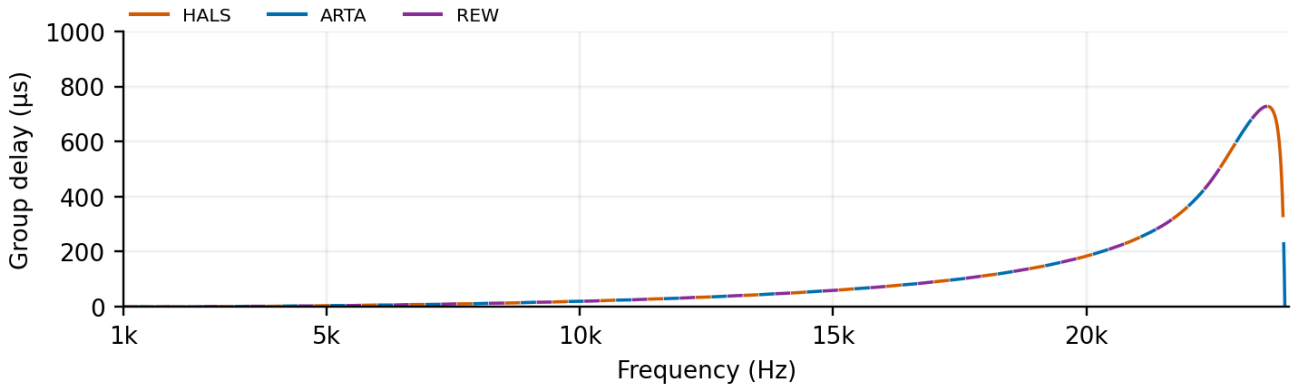
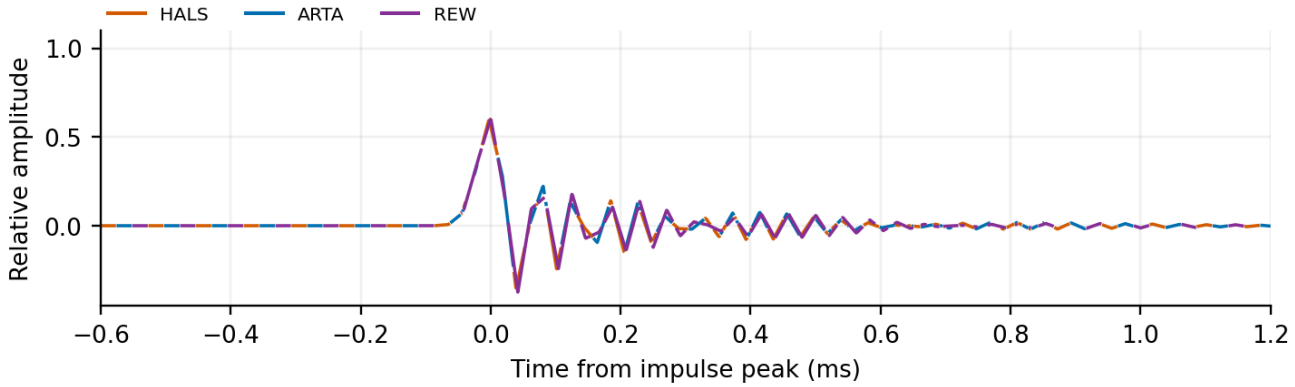


Figure 8. Impulse Response



The three engines recover closely matching hardware responses. The low-frequency roll-off is consistent with AC coupling, while converter reconstruction and anti-alias filtering affect the high-frequency magnitude, phase and group delay.

The response is -0.35 dB at 20 Hz and -0.41 dB at 20 kHz (Figure 5). At 20 kHz HALS measures -277° of phase (Figure 6) and 184 µs of group delay (Figure 7) relative to 1 kHz.

This interface behaviour can be acceptable for much acoustic work, but can contribute to apparent high-frequency decay tails in CSD and wavelet analysis due to the HF group delay. Oscillation around the impulse peak (Figure 8) is an expected time-domain consequence of the bandwidth and phase response; oscillation alone does not establish a corrupt or poor capture, despite the significant visual impact.

MOTU M4 hardware loopback with interface correction

HALS interface correction is compared with ARTA and REW's interface correction. ARTA's optional filter for dual-channel IRs is disabled in this section.

Figure 9. Frequency Response

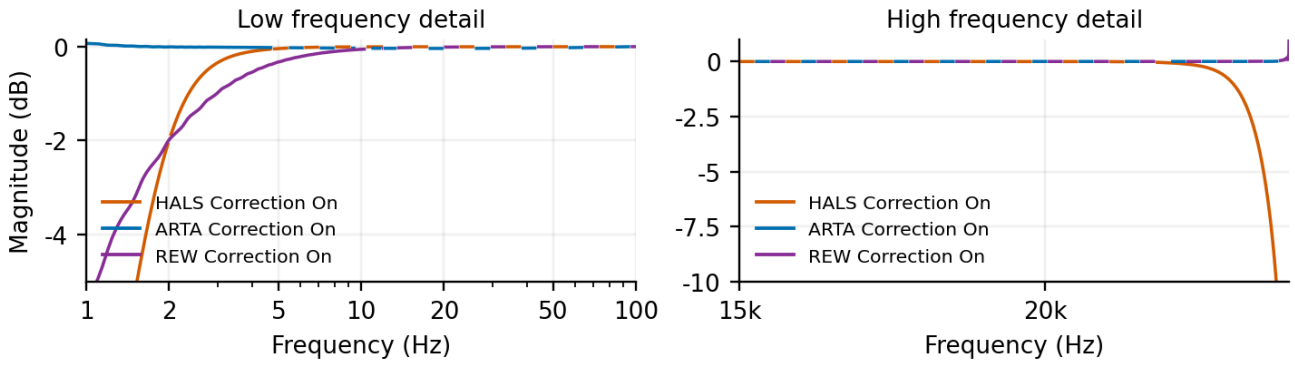


Figure 10. Phase Response

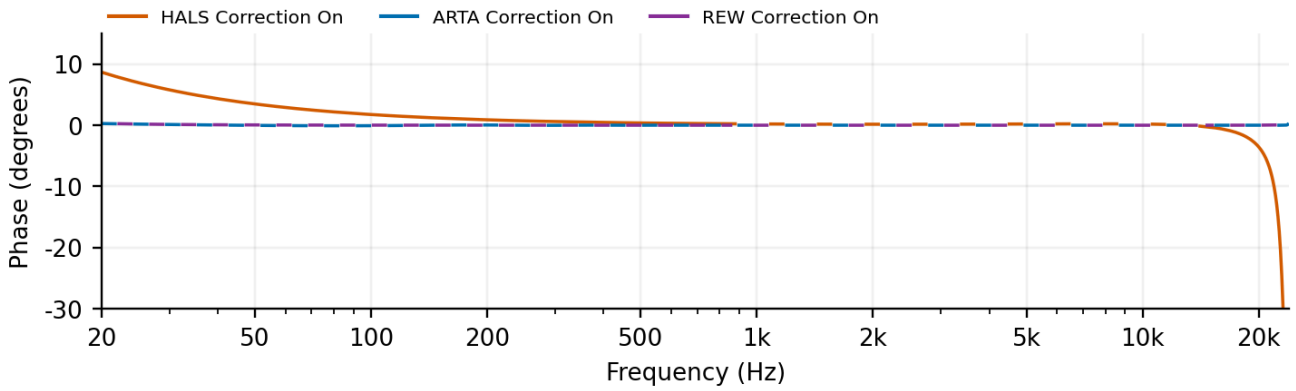


Figure 11. Group Delay

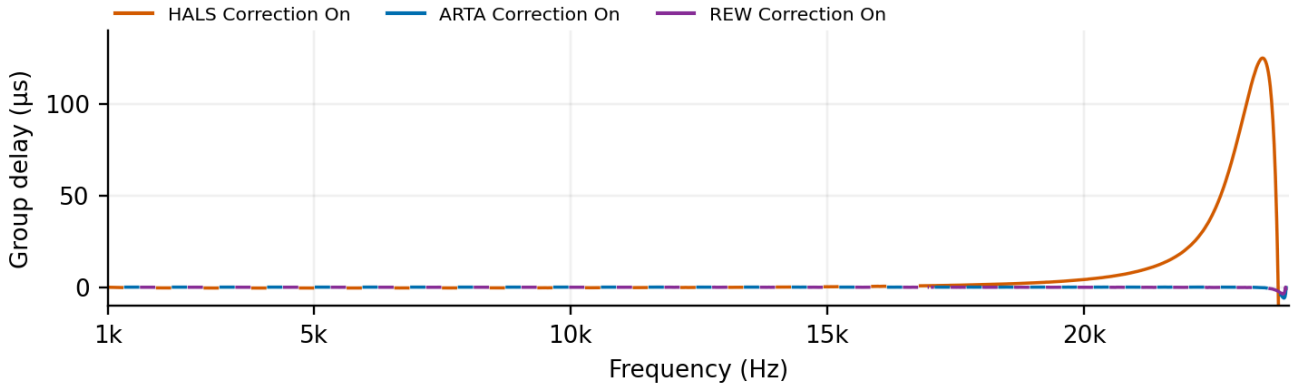
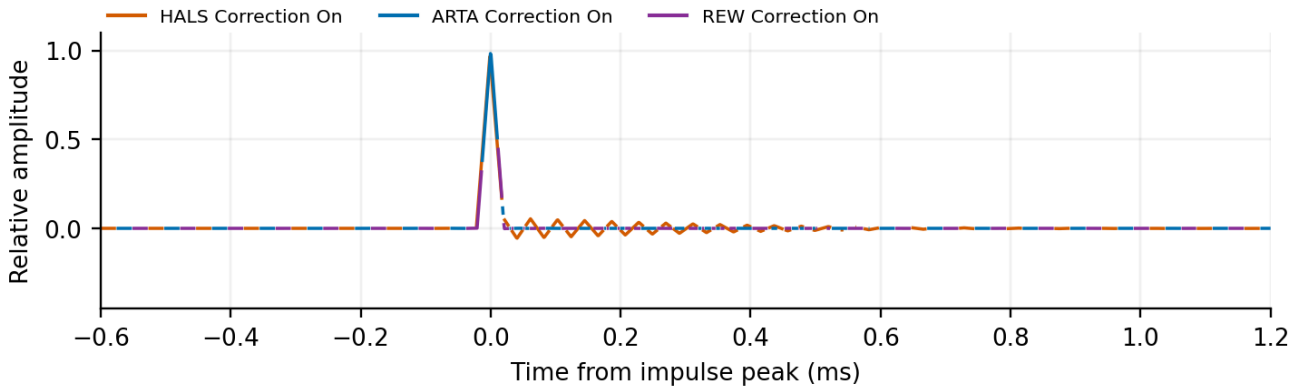


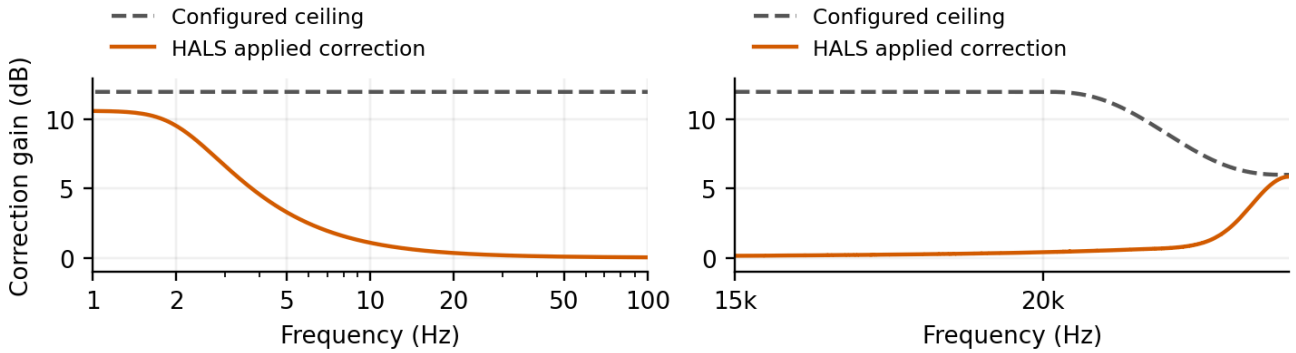
Figure 12. Impulse Response



Interface correction substantially reduces the hardware's contribution through the useful passband. At 20 kHz HALS frequency response is 0.00 dB relative to midband (Figure 9), with -3.6° of phase (Figure 10) and 4.2 µs of group delay (Figure 11) relative to 1 kHz. The impulse response shows that post-peak ripple is significantly reduced (Figure 12). ARTA and REW remain closer to ideal at the very top end because they apply stronger inverse correction.

Correction gain

Figure 13. Correction Gain



HALS deliberately limits correction gain to 12 dB through 20 kHz, tapering to 6 dB at Nyquist (Figure 13). Since corrective boost also amplifies recorded input noise, this limit represents a compromise between correcting the interface response and avoiding excessive noise amplification at the bandwidth extremes.

In a clean electrical loopback with >100dB SNR, the stronger correction applied by ARTA and REW can appear to produce a superior result. However, HALS prioritises noise performance in practical acoustic measurements, accepting small residual response errors where the required correction exceeds the conservative gain limit.

HALS also shapes the residual bandwidth roll-offs to be minimum phase. This avoids introducing pre-ringing, at the cost of small residual phase and group delay deviations near the bandwidth limits.

MOTU M4 interface correction with bandwidth filtering

ARTA offers an optional filter for dual-channel impulse responses to suppress noise amplified by interface correction. It attenuates the HF edge with linear-phase filtering which retains the corrected phase and group delay. This section compares that mode with HALS's gain-limited minimum-phase residual. REW does not provide an equivalent HF noise-suppression filter in its standard measurement settings, so it is excluded from this section.

Figure 14. Frequency Response

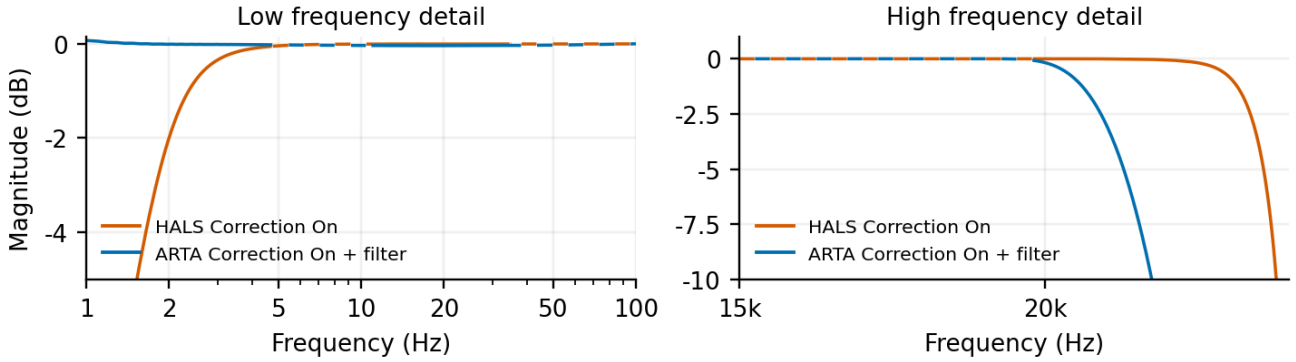


Figure 15. Phase Response

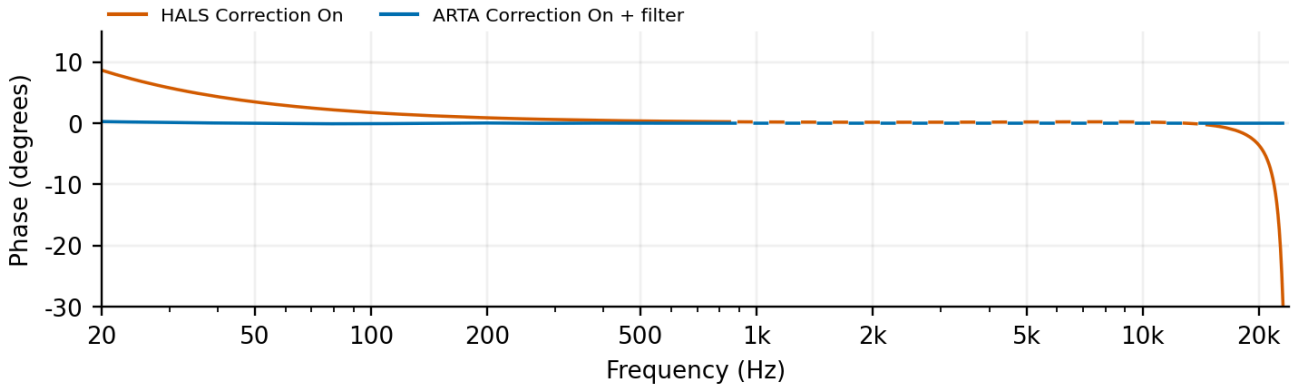


Figure 16. Group Delay

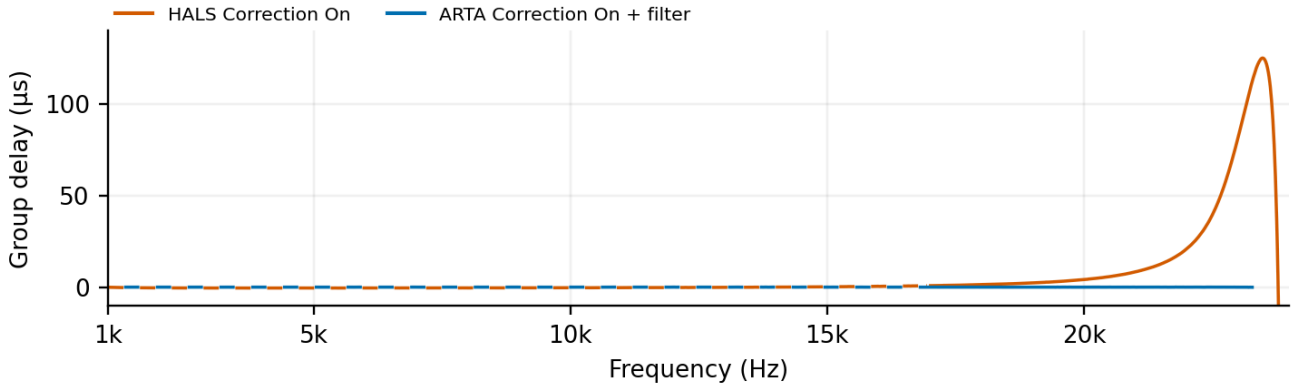
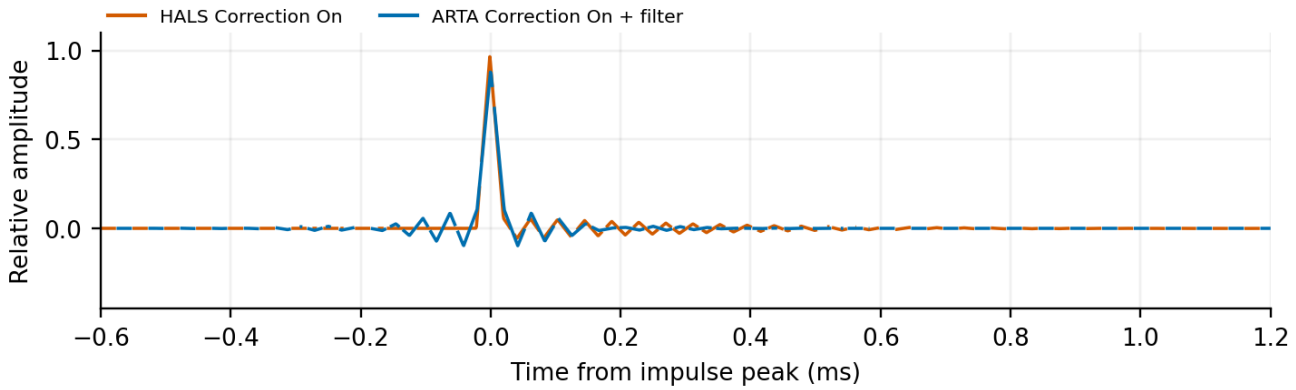


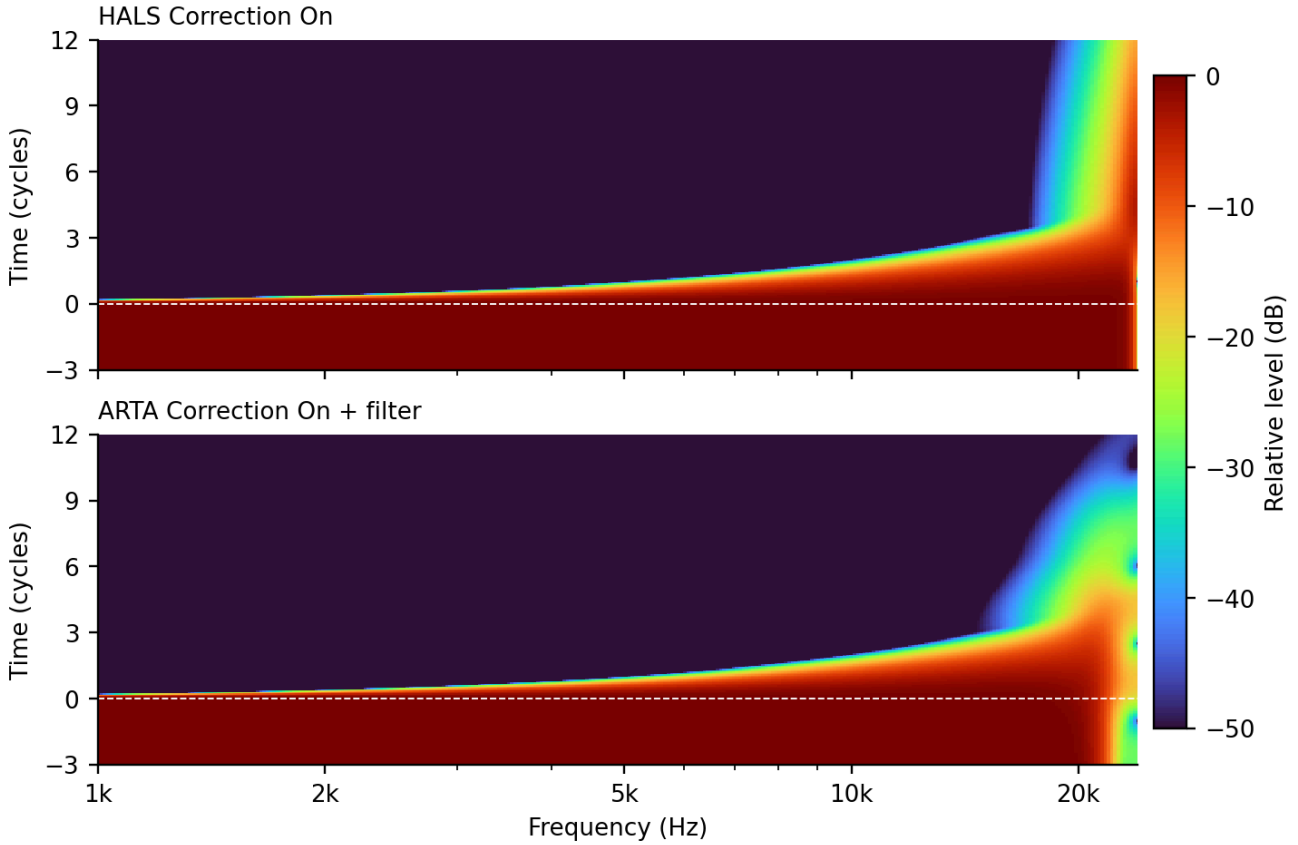
Figure 17. Impulse Response



ARTA's optional filter rolls off the frequency response close to Nyquist (Figure 14), while leaving the corrected phase (Figure 15) and group delay (Figure 16) nearly unchanged. The impulse waveform shows the pre-peak ringing introduced by this approach (Figure 17). HALS avoids this added pre-ringing by retaining a minimum-phase residual.

Decay analysis of the corrected impulses

Figure 18. CSD



The flatter group delay of ARTA's approach (Figure 16) might suggest that CSD plots would have shorter HF decay tails than HALS. At 20 kHz, the time to decay by 6 dB is approximately 110 µs for HALS and 122 µs for ARTA with its filter enabled (Figure 18). This is a small difference; neither approach has a substantial decay advantage at 20 kHz in this comparison.

The explanation lies in how CSD and burst-decay plots interact with the different time extents of ARTA's zero-phase filtering and HALS's minimum-phase residual. HALS avoids added pre-ringing and gives predictable behaviour when users exclude pre-peak time. The plots use -3 to +12 cycles and a 0 to -50 dB level range. Time zero is the main impulse peak.

MOTU M4 interface correction at 96 kHz

Where minimal audio-band group delay and HF decay are required for analysis, capturing at a higher sample rate such as 96 kHz moves the bandwidth edge away from the audio band. These measurements use 96 kHz with the same 5.461 s sweep duration. ARTA's dual-channel filter remains enabled. Nyquist is now 48 kHz.

The frequency response (Figure 19), phase (Figure 20), group delay (Figure 21) and impulse waveform (Figure 22) show the result at 96 kHz. The wider bandwidth moves the steep roll-off away from 20 kHz. If response or decay above 20 kHz is important, a higher sample rate provides a more useful measurement bandwidth.

Figure 19. Frequency Response

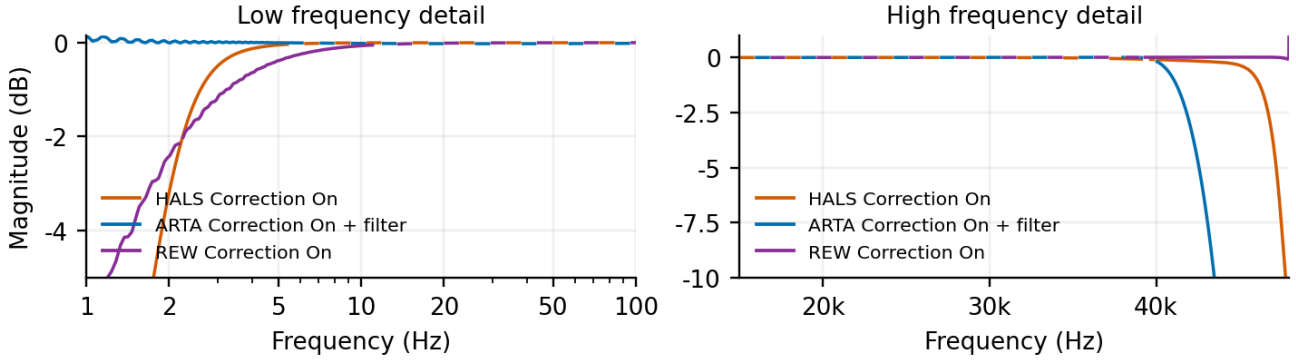


Figure 20. Phase Response

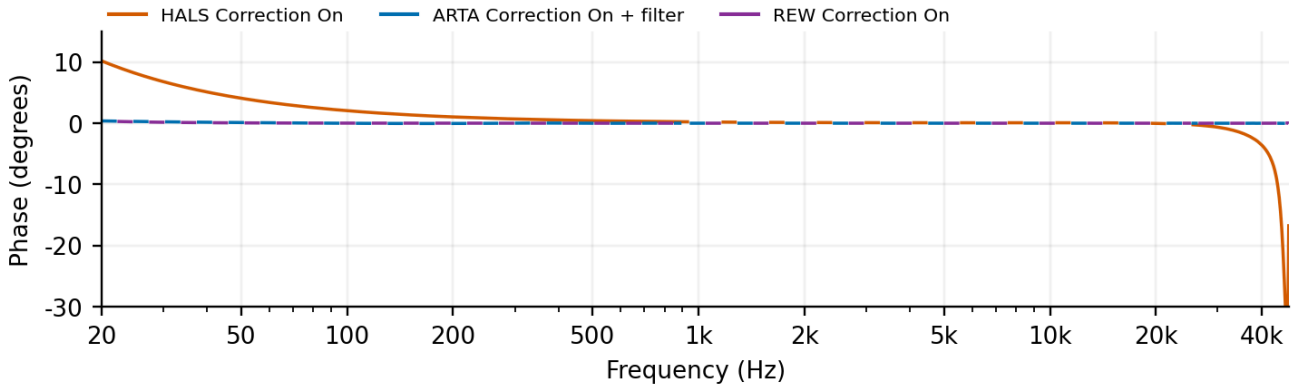


Figure 21. Group Delay

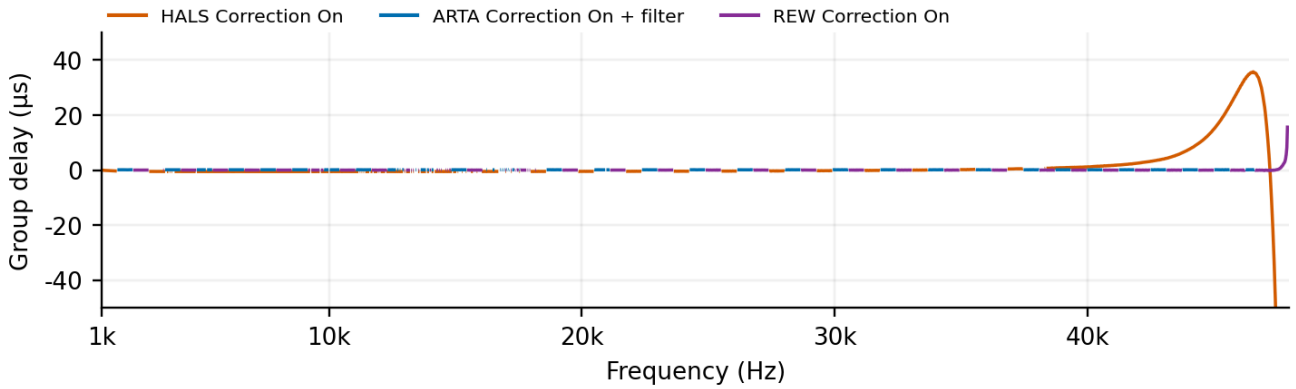
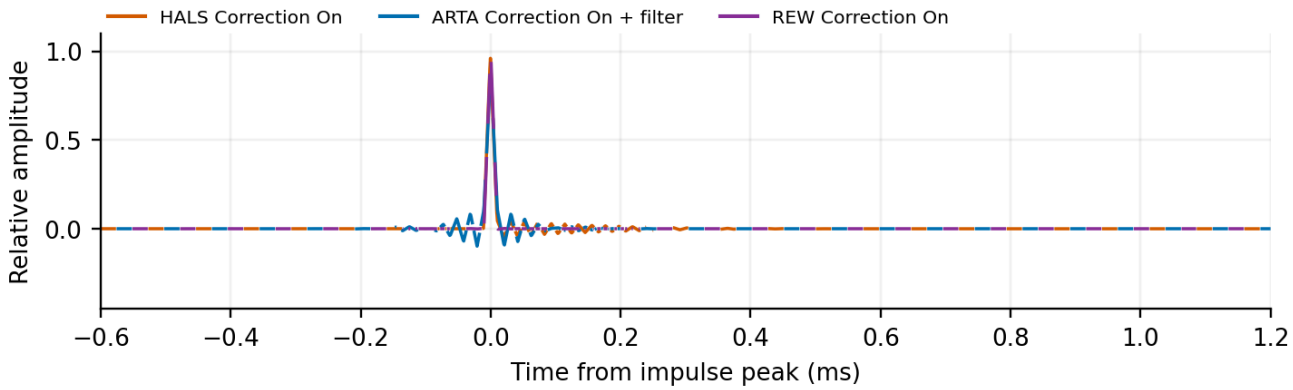
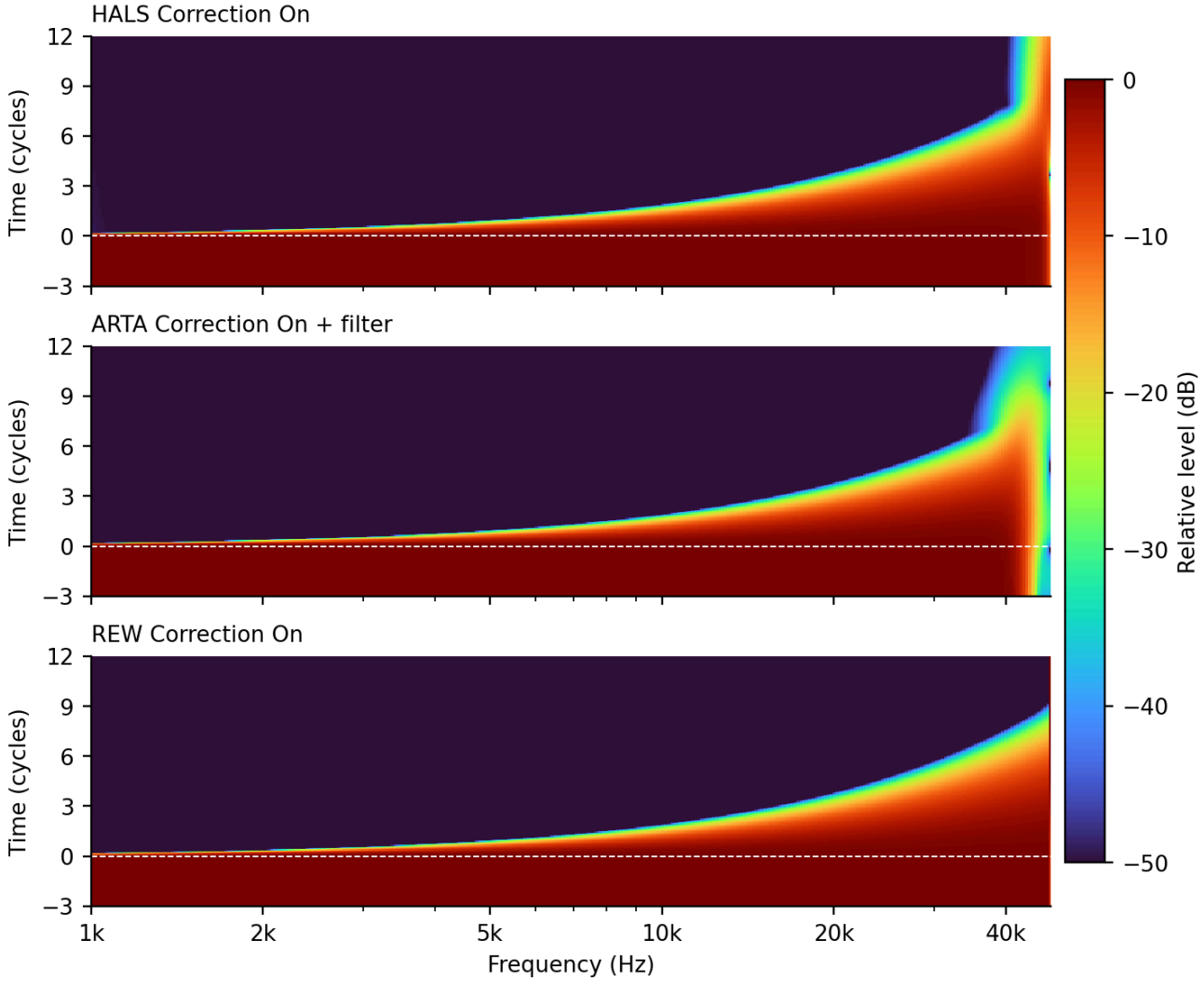


Figure 22. Impulse Response



Decay analysis at 96 kHz

Figure 23. CSD



The wider frequency range makes the benefit of 96 kHz visible: 20 kHz lies well inside the available bandwidth (Figure 23). These sonograms use the same cycle and level ranges as Figure 18.

The digital and hardware loopbacks establish accurate HALS impulse-response recovery and close agreement with established measurement software. Its gain-limited correction substantially reduces interface effects while avoiding excessive noise boost and retains a causal minimum-phase impulse.